

Ilari Angervuori

Highly Skilled in Stochastic Performance Optimization and Trade-Off Analysis of Satellite Communications

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Profile

A mathematically highly skilled R&D engineer in satellite communications with a pioneering publication record in stochastic geometry modeling of satellite communication systems, including a first-author publication in the high-impact journal Institute of Electrical and Electronics Engineers (IEEE) Transactions on Communications (TCOM) and a first-author presentation in the flagship conference IEEE GLOBECOM. A creative mindset that seeks simplicity, not complexity, and is not afraid to fiddle with the details. Committed to contributing to the growing industry of satellite systems and to bringing in my unique expertise in a collaborative, challenging, development-driven environment. Always striving to grow professionally as well as spiritually, and to cultivate compassion and humility in all aspects of life. FI/EU citizen.

Skills

• Data analysis and visualization • Inter-institute collaboration • Research • Stakeholder management • Writing
• 3GPP NR NTN specs. • C, MATLAB, PYTHON • Interference estimation in a LEO uplink • NTN PHY and MAC simulations, and performance and trade-off analysis • Stochastic processes

Experience and Impact

- 5/2025- **Senior Researcher**, *Aalto University*, School of Electrical Engineering, Espoo, Finland
- Managed 2 R&D projects as the corresponding author and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
 - Mentored junior researchers as a bachelor's degree supervisor and helped to establish future talents.
 - Voluntary work as a peer reviewer in various journals, including IEEE TCOM, IEEE TWC, IEEE WCL, IEEE COMML, and IEEE TMC.
 - Proposed a novel closed-form Gaussian autocorrelation function for the aggregate interference at a LEO satellite terminal from Poisson-distributed terrestrial signal terminals, applicable, *e.g.*, in machine learning (ML) algorithms for satellite communications as a tractable hypothesis class.
 - Proposed, analyzed, and simulated successive interference cancellation (SIC) algorithms that improved coverage performance by 900% for the third-strongest Earth transmitter in terms of signal-to-interference ratio (SIR). The SIC also improves user fairness by serving multiple terrestrial terminals per cell, while preserving the average total network throughput at the LEO satellites, which is optimized around 1 (bit/s/Hz).
- 8/2023- **Research Internship**, *University of Notre Dame*, Dept. of Electrical Engineering, South Bend, IN, US
- 2/2024
- Based on rigorous mathematical analysis, a novel statistical SIR model was proposed by characterizing the SIR at a LEO satellite receiver using a Lomax distribution: a well-known heavy-tailed distribution. Consequently, many results for the Lomax distribution can be exploited in performance analysis and optimization of LEO communications.
 - Coordinated a distributed team in a cross-functional project and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in the high-impact journal IEEE TCOM.
 - Identified an optimal cell size in a LEO uplink that maximizes average throughput and quantified a trade-off between average throughput and performance consistency: -10 dB cell size means high average throughput but highly varying link quality of service (QoS) over the beams, whereas -3 dB cells entail more stable QoS but degraded throughput.
 - International relations.
 - Secured project financing of 4349 € for personal internship expenses from the Väisälä Foundation.
- 05/2021- **PhD Researcher**, *Aalto University*, Signal Processing of Wireless Networks, Espoo, Finland
- 05/2025
- Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, MATHEMATICA, OCTAVE, PYTHON) including algorithms for analysis, simulation, and visualization, used across several research projects.
 - Developed a novel and well-cited stochastic geometry framework for the analysis of LEO networks, reducing simulation time by an order of magnitude 100 times more computationally efficient than the MATLAB Satellite Communication Toolbox, while providing equivalent results on the network performance metrics.
 - Industry presentation at IEEE GLOBECOM on a novel stochastic geometry system model for LEO uplink performance metrics, characterizing the SIR with a gamma distribution. Consequently, many results for the gamma distribution can be exploited in performance analysis and optimization of LEO communications.
 - Proposed, analyzed, and simulated a minimal-overhead slotted ALOHA-like ISL data routing algorithm in a constellation of 10 planes, 6 satellites in each plane at the orbital altitudes 1000 km, using graph algorithms. The QoS was optimized at an average latency of 30 ms and total network spectral efficiency 1 (bit/s/Hz).
 - Proposed, analyzed, and simulated a minimal-overhead, grant-free, random MAC for LEO uplink that mitigates the high variance in link performance during a use period while preserving the average network throughput, which is optimized around 1 (bit/s/Hz).

Education

Expected 2026 **Doctor of Science, PhD**, *Aalto University, School of Electrical Engineering*, Signal Processing for Communications, • Stochastic Geometry Analysis and Simulation of LEO networks
MSc, *Univ. of Helsinki*, Applied Analysis, • Finite Element Method (FEM) • Antenna Theory
BSc, *Univ. of Helsinki*, Applied Analysis, • Optimal Control Theory • System Performance Optimization

References

- Prof. Martin Haenggi, University of Notre Dame, mhaenggi@nd.edu, +1 574-631-6103
- Prof. Risto Wichman, Aalto University, risto.wichman@aalto.fi, +358 40 0800801
- MSc. Abid Afridi, Aalto University, abid.afridi@aalto.fi